THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum of the first 12 terms of an AP is 510 and its first term is 4. The common difference is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 9

2. The exterior angle at D of cyclic quadrilateral $ABCD$ (side CD produced) equals the interior angle at B, which is 72°. The interior angle $\angle ADC$ is: 1

(a) 118° **(b)** 98° **(c)** 128° **(d)** 108°

3. The area of a rhombus of side 13 cm and one diagonal 10 cm is (cm^2): 1

(a) 120 **(b)** 122 **(c)** 121 **(d)** 119

4. A bag has 36 balls, red or blue. If $P(\text{red}) = \frac{5}{12}$, the number of blue balls is: 1

(a) 22 **(b)** 20 **(c)** 21 **(d)** 23

5. Three numbers in AP have sum 30 and product 840 (with $d > 0$). The largest number is: 1

(a) 18 **(b)** 14 **(c)** 15 **(d)** 10

6. AB is a diameter, C and D are on the circle (same side of AB) and $ABCD$ is cyclic with $\angle DAB = 48^\circ$. Then $\angle BCD =$ 1

(a) 122° **(b)** 142° **(c)** 152° **(d)** 132°

7. A triangle has sides 10 cm, 17 cm, 21 cm. The length of the shortest altitude (cm) is: 1

(a) 8 **(b)** 10 **(c)** 7 **(d)** 9

8. A letter is chosen from the word MISSISSIPPI. The probability that it is not the letter I is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{8}{11}$ **(d)** $\frac{7}{11}$

9. A town's population is 10000 and grows by 10% each year. After 2 years it is: 1

(a) 12000 **(b)** 12100 **(c)** 12300 **(d)** 12200

10. A chord AB of length 14 cm lies 24 cm from the centre O. The area of triangle OAB (cm^2) is: 1

(a) 170 **(b)** 167 **(c)** 169 **(d)** 168

11. A rope of length 88 cm (take $\pi = \frac{22}{7}$) forms a circle. The same rope forms a square. The area of the square (cm^2) is: 1

(a) 484 **(b)** 485 **(c)** 483 **(d)** 486

12. The probability that a seed germinates is $\frac{7}{10}$. Out of 2000 seeds the expected number that will NOT germinate is: 1

(a) 602 **(b)** 599 **(c)** 601 **(d)** 600

13. The first term of a GP is 3 and the 8th term is 384 (with $r > 0$). The common ratio is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 1

14. If two chords of a circle bisect each other, they are: 1

(a) both diameters **(b)** perpendicular to each other
(c) parallel **(d)** unequal

15. If the radius of a circle is doubled, its area becomes: 1

(a) eight times **(b)** two times **(c)** three times **(d)** four times

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A circle of radius 7 cm is inscribed in a square (take $\pi = \frac{22}{7}$). Find the area of the square left uncovered by the circle. 2

17. A bag has 12 red balls and some blue balls. If $P(\text{blue}) = \frac{1}{4}$, find the number of blue balls. 2

18. Which term of the AP $5, 11, 17, \ldots$ is the first term to exceed 200? 2

19. $ABCD$ is a cyclic quadrilateral and $\angle ABD = 32^\circ$, $\angle DBC = 41^\circ$. Find $\angle ABC$ and $\angle ADC$. 2

20. The area of a rhombus is 120 cm^2 and one diagonal is 10 cm. Find the other diagonal and the side of the rhombus. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the sum of all natural numbers between 10 and 200 that leave remainder 2 when divided by 5. 3

22. AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 34^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. 3

23. The sides of a triangle are 13 cm, 14 cm and 15 cm. Find its area and the altitudes on the sides 13 cm and 14 cm. 3

24. A letter is chosen at random from the word MISSISSIPPI. Find the probability that it is 3
(i) not S,
(ii) a vowel or S.

OR

Three positive numbers are in GP. Their product is 216 and the sum of the first and third is 15. Find the numbers. 3

25. Two circles of radii 13 cm and 20 cm intersect in a common chord of length 24 cm. Find the distance between their centres if they are on 3
(i) opposite sides,
(ii) the same side of the common chord.

OR

A rope 88 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A circle of radius 13 cm has a chord AB of length 10 cm. Find the distance of AB from the centre O, the area of $\triangle OAB$ and the perimeter of $\triangle OAB$. **5**

OR

A park is a rectangle 60 m $\times$ 40 m. A circular flower bed of radius 7 m (take $\pi = \frac{22}{7}$) is in it, and a path 5 m wide runs round the park inside its boundary. Find the area of the park excluding the path, the area of the grass if the bed is removed from this, and the cost of turfing the grass at ₹3 per m^2. **5**

27. A bag has 7 red, 8 blue and 12 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{4}$. How many green balls were removed? **5**

OR

A ball is dropped from 81 m and rebounds to 2/3 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. **5**

END OF QUESTION PAPER — Total: 50 Marks

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